Pivotum

THE DEGREE RISK INDEX · FALL 2026 EDITION

Is an Engineering Degree Safe From AI?

The complete profile — the full scores by track, the six factors behind them, the honest downsides, and what to ask before you commit.

3.4

AI EXPOSURE · CIVIL / STRUCTURAL (PE, SITE-BASED)

where 10 is most at risk

Read this one first; send the student version to them.

HIGHLIGHTING IS COLOUR-CODED

THE FINDING EXPOSURE PROTECTION METHOD & HONESTY

Before you read this — a note for parents

You're likely reading this before your child is. That's normal, and it's worth pausing on how you use it.

Don't lead with the scores. If you open the conversation with a risk number, they either get frightened and shut down, or defensive about a decision they've already made.

The better approach: read this yourself first, then share it as *information you both get to react to*.

What this is. A facts document, not a recommendation.

Three things to keep in mind:

1. **They're choosing, not you.** 2. **This is a starting point, not a ruling.** 3. **Don't do it all at once.**

A practical note: there's a separate short version written directly to them. Send them that one.

And one thing specific to engineering. This is the most encouraging profile in the index, and that creates its own risk. A document that says "this looks good" can shut down thinking rather than open it up. The scores here are strong, and the honest difficulties — the length of licensure, the cost of the degree, whether the work actually suits them — haven't gone anywhere. Use the good news to open the conversation, not to close it.

The short answer

Yes — and for a technically-minded student, the engineering-versus-computer-science comparison deserves far more thought than it usually gets.

Licensed, site-based engineering scores **4.0** out of 10. Desk-based design and analysis scores **6.3**. Entry-level software development, for the same kind of mind and much the same maths, scores **8.1**.

And the labour markets point the same way. There are roughly three open engineering positions for every qualified candidate, and nearly half of working US engineers are aged 50 or over. Recent computer science graduates face 6.1% unemployment.

Same aptitude. Opposite entry markets.

The scores

DISCIPLINE	2023	2025	FALL 2026	3-YR MOVEMENT	BAND
Electrical power & grid	3.3	3.6	**3.9**	+0.6	Low–Moderate
Civil / structural (PE, site-based)	3.4	3.7	**4.0**	+0.6	Low–Moderate
Mechanical (industrial)	3.9	4.1	**4.4**	+0.5	Moderate
Manufacturing / process	4.4	4.7	**4.9**	+0.5	Moderate
Design & analysis (desk, CAD-centred)	5.2	5.8	**6.3**	+1.1	Moderate–High

Scores run 1–10, *where* 10 *is most at risk*. *For context: the median profession in this edition sits around* 5.5, *a service electrician scores* 2.5, *bedside nursing* 2.8, *senior software architecture* 5.4, *and entry-level software development* 8.1.

Read the table this way: the spread runs almost exactly along the line between the site and the office. Everything that involves being somewhere physical, with a stamp and a liability attached, sits below 5. Everything that happens at a desk climbs.

Engineering versus computer science — the comparison worth making

This is the most decision-relevant table in the index for a student who is good at maths and likes building things.

	SOFTWARE	ENGINEERING
Regulatory protection	None — no licence exists	Strong — the PE stamp is a legal monopoly
Physical requirement	None	Real — sites, commissioning, inspection
Accountability for catastrophic failure	Commercial	Legal and personal
Entry route	Market-driven, contracted 67% in one year	Licensure requires supervised experience years
Graduate market	6.1% unemployment	~3 open roles per qualified candidate
Entry-level score	**8.1**	**4.0** licensed / 6.3 desk
Starting salary	~$87,000	~$81,198 average across disciplines

Software has two of our six protection factors sitting at essentially zero for everyone in the profession, permanently. Engineering has neither of those gaps at the licensed end.

The salary gap at entry is real and it is smaller than most people assume — about $6,000. Software pulls ahead at mid-career. But it pulls ahead *for the people who get in*, and that is precisely the part that has changed.

Why licensed engineering scores low — the six factors

Here's how civil and structural engineering answers them.

HOW MUCH OF THIS JOB CAN AI ALREADY DO? (35% OF THE SCORE)

A substantial and growing share of the desk work.

CAD drafting and documentation. Standard-case calculation. Simulation setup and iteration. Generative design across cost, material and structural constraints. Compliance and code checking.

Each of those was, until recently, a junior engineer's assignment. Generative design tools now produce and compare dozens of options against constraints in the time it once took to draw one.

What resists: the stamp and the liability behind it, site work, commissioning, and judgment about whether an analysis actually applies *here*.

HOW HARD WILL IT BE TO LAND THAT FIRST JOB? (15%)

Among the better on-ramps we measure, and for two independent reasons.

The first is structural: the PE licence requires supervised experience years, so firms have an institutional reason to employ and train juniors — the same mechanism that protects nursing and accounting.

The second is demographic and unusually strong. Roughly three open engineering positions exist per qualified candidate. Nearly half of working US engineers are 50 or older, and retirements are outpacing new graduates in several disciplines.

DOES IT HAVE TO BE DONE IN PERSON, WITH YOUR HANDS? (15%)

For the licensed, site-based disciplines, substantially.

Site visits, commissioning, inspection, snagging, and the judgment that comes from standing in the actual conditions. This is where the internal spread begins — a site-based civil engineer and a desk-based analyst do very different jobs with the same degree.

DOES SOMEONE NEED A HUMAN THEY CAN TRUST AND HOLD RESPONSIBLE? (15%)

Yes, and with a severity few professions match.

When a structure fails, people die and someone is legally answerable. That accountability attaches to a named person with a licence, and no advance in capability transfers it.

DOES THE LAW REQUIRE A LICENSED HUMAN? (10%)

For anything that must not fail, comprehensively.

The PE stamp is a legal monopoly. Certain designs must be signed by a licensed engineer, and personal liability follows the signature. Licensing moves at the speed of legislation and public safety, not at the speed of technology.

HOW OFTEN DOES THE JOB HIT GENUINELY NEW, HIGH-STAKES SITUATIONS? (10%)

Constantly — and for a reason worth understanding precisely.

It is not that the calculations are hard. Calculations are exactly what machines do well. It is that every site is different and the consequences of being wrong are structural. A bridge sits on particular ground, in a particular seismic and thermal environment, under a particular jurisdiction's code, built by a particular contractor. The combination has no precedent even where every individual element does.

The question the profession hasn't answered

This deserves its own section, because it is the one place we think engineering's protection may be quietly weakening.

The PE licence mandates logged experience years under a licensed practitioner. That props the on-ramp open — it is the same mechanism that protects nursing, and it is genuinely valuable.

But what happens *inside* those years has changed.

A candidate who spends them directing generative tools and reviewing output accumulates the same hours as one who spent them drawing by hand and running calculations. Are they building the same judgment?

Licensure certifies that someone has done the time. It assumes the time teaches. If the content of those years shifts from producing work to supervising work, the assumption may stop holding — and the licence would continue certifying something it no longer measures.

This sorts the professions in a way worth carrying to other careers. In nursing, medicine and the trades, mandated experience hours are hands-on by necessity — you cannot supervise your way through clinical hours or an apprenticeship. In engineering, architecture and accounting, the mandated hours are desk-based, and desk work is exactly what is changing.

We rate the protection 9.0 because it is currently doing its job. We flag it as the factor most likely to be hollowed out from the inside rather than repealed.

What the trend shows

Licensed site-based: 3.4 → 3.7 → 4.0. Up 0.6 in three years.

Desk-based design and analysis: 5.2 → 5.8 → 6.3. Up 1.1 — nearly double the pace.

The gap widened from 1.8 points to 2.3. The pattern is now familiar across this index: the physical, accountable end holds while the screen-based end drifts. What is different in engineering is that both ends started from a good position, so even the drifting end remains below the index median.

How durable is the protection?

PROTECTION	STRENGTH TODAY	DURABILITY	WHERE THE PRESSURE IS COMING FROM
Regulatory / licensing	Strong	**Most durable**	Moves at the speed of legislation and public safety. But see section 5 on logged hours.
Human trust / accountability	Strong	**Durable**	Someone must be legally answerable when a structure fails.
Judgment under novelty	Strong	**Eroding slowly**	Standard-case analysis is automating; site-specific judgment is not.
Physical / embodied	Real for site disciplines	**Most fragile long-term**	The protection physical AI directly targets — though construction sites are among the hardest environments for robotics.

The honest read for a seventeen-year-old: they will practise for roughly forty years. The two protections most likely to still be standing are the licence and the liability, because both are rooted in law rather than in what technology can do.

How the role will actually change

WAS THE JOB	IS BECOMING THE JOB
Drawing it	Specifying it, then checking what was generated
Running the standard calculation	Judging whether the standard case applies here
Producing one design option	Choosing among fifty generated options
Learning by drafting for two years	Learning by reviewing — see section 5
Compliance checking by hand	Verifying automated compliance output

One consequence worth naming. The value moves upstream, toward specifying the problem and judging the answer, and away from producing the artefact. That is good news for engineers who want to be responsible for outcomes, and less good for anyone who chose engineering because they enjoy the drawing.

Human strengths that will matter most

1. **Site judgment** — reading actual conditions against what the drawings claim. The least

> automatable thing in the profession.

2. **Accountability** — carrying the stamp and the liability. 3. **Judgment on one-off conditions** — knowing when the standard case doesn't apply. 4. **Coordinating people who disagree** — contractors, clients, regulators, other disciplines. 5. **Verification** — knowing when a generated design or analysis is confidently wrong.

Notably absent: drafting speed, software fluency in any particular CAD package, and volume of standard calculations. These were the traditional markers of a strong graduate engineer and are the fastest-depreciating.

Other things to consider about this job

What's genuinely good about it

The market is exceptionally strong, and it is strong for demographic reasons that will outlast any technology cycle.

- **Roughly three open positions per qualified candidate**, with 76% of employers in one survey reporting difficulty finding qualified candidates
- **Nearly half of working US engineers are 50 or over** — retirements are outpacing new graduates in several disciplines
- **Average starting salary of $81,198**, up from $78,731 the previous year
- **BLS median wages:** civil $98,540, mechanical $99,510, electrical $111,150
- **Job growth through 2034:** industrial 11%, mechanical 9.1%, electrical 7.2%, civil 7%

And the PE licence is unusually good value. Industry salary surveys consistently show licensed engineers earning 10–25% more than unlicensed peers at the same experience level — a lifetime premium estimated between $300,000 and $700,000. Very few professional qualifications have a return that clear.

What engineers report as rewarding: building things that exist afterwards, working at a scale individuals cannot reach alone, the intellectual satisfaction of a hard constraint problem, and — distinctively — the weight of being trusted with public safety.

What's genuinely hard about it

The degree is demanding and attrition is real. Engineering has among the higher switch-out rates of any major, concentrated in the first two years of maths-heavy foundational content.

Licensure takes years. Typically four years of supervised experience after graduating, plus examinations. Career-defining, and slow.

The work can be less creative than students expect. A great deal of professional engineering is codes, standards, documentation and coordination rather than design.

Site-based roles can mean travel and relocation, particularly early on and particularly in civil.

Who this work suits — and who it doesn't

Engineering tends to suit people who:

- **Are comfortable with responsibility for physical consequences.** The defining trait.
- **Like constraints.** Engineering is problem-solving inside hard limits — cost, material, code, physics — rather than open-ended creation.
- **Are precise and can be checked.** The work is reviewed, stamped and audited.
- **Want to see the thing exist afterwards**

It's harder for people who:

- Want **creative latitude** — architecture is closer to that than engineering
- Dislike **regulation and documentation**, which is a large share of the job
- Struggle with **sustained maths-heavy foundations**, which is where most attrition happens

What else is moving this market

Electrical engineering has an extraordinary tailwind, and it comes from AI itself. Data centre construction is running at record pace. Lawrence Berkeley National Laboratory estimates AI data centres could consume between 325 and 580 terawatt-hours by 2028 — **6 to 12% of total US electricity consumption.**

That makes electrical engineering possibly the single most demand-secure choice in this entire index: a career protected from AI that is simultaneously being pulled by it.

Infrastructure investment is elevated across transport, water and grid in most developed economies, and civil demand follows it.

One honest caution: some of the current construction and data-centre demand is project-driven and will end when the builds do. The demographic case — an ageing workforce retiring faster than replacements arrive — holds regardless of the cycle. A student should weight the demographics more heavily than the current boom.

The AI-native advantage — what to actually do about it

The situation: generative design, automated compliance checking and simulation tooling are already standard in large practices. The people who can direct and verify them are not.

WHAT AN AI-NATIVE ENGINEER LOOKS LIKE

1. Verify. Knowing when a generated design is subtly wrong — buildable on screen and not in reality. This requires understanding the physical constraints well enough to catch a plausible error, which is why the fundamentals matter more rather than less.

2. Specify. Framing the constraint set precisely enough that generated options are usable. The skill has moved upstream.

3. Judge applicability. Knowing whether a standard case fits these ground conditions, this contractor, this jurisdiction. The core of professional judgment.

4. Own the stamp. Everything in this profile points at accountability. Get on the licensure track early and deliberately.

5. Bridge to site. Being the person who can connect what the model says with what the site actually shows.

CONCRETE PREPARATION

Before the degree: understand that the discipline choice matters more than the school. Electrical and civil are in the strongest demand positions; the desk-versus-site distinction matters more than any ranking.

During: take the co-op or placement year — it is the single best predictor of outcomes, and engineering programmes are unusually good at providing it. Get site exposure specifically, not just office placement. Treat the maths-heavy foundations as the thing verification will later depend on.

After: start the PE track immediately. The four supervised years are the protection, and every year of delay is a year of the premium foregone.

The routes in

ROUTE	NOTES
Accredited engineering degree	The standard route. Accreditation matters for licensure — check it.
Degree apprenticeship	Expanding, paid, and combines the degree with the supervised experience. Seriously underused.
Engineering technician → degree	Slower, but earns throughout and arrives with site experience.
Discipline choice within the degree	Matters more than school choice. Electrical and civil are in the strongest positions.

Where an engineering degree can take them later

Broad and well-regarded: professional practice across disciplines, project and construction management, energy and infrastructure, manufacturing and process, technical sales, consulting, regulation and standards, and engineering leadership.

The pattern that runs through this index applies: the further a role moves from physical accountability, the higher its exposure climbs. Desk-based analysis at 6.3 is the warning inside the profession itself.

What to look for in an engineering program

1. Accreditation. Non-negotiable if licensure is the goal. Check it directly rather than assuming.

2. Co-op or placement — and whether it includes site exposure. Office placement is good; site placement is better and rarer.

3. Does the curriculum teach generative design tools *and* their failure modes?

4. PE preparation. Does the programme actively support the licensure pathway, or leave it to graduates to work out?

5. Discipline strength. A department strong in electrical or civil is worth more than a generally-ranked institution.

Questions to ask an engineering program

On accreditation and licensure:

- Is the programme accredited for professional licensure in the states where graduates practise?
- What proportion of graduates are on the PE track five years out?

On placement:

- What proportion complete a co-op or placement, and is it guaranteed?
- How many include genuine site experience rather than office-based work?

On the curriculum:

- How does the programme teach students to work with generative design and simulation tools?
- Where do students learn to identify when generated output is wrong?
- What changed in the last two years?

On outcomes:

- What proportion are in engineering roles at six months, and in what settings?
- What's the attrition rate in years one and two, and why do students leave?

Red flags: unclear accreditation; placement described as "available"; no PE pathway support; a curriculum unchanged since 2022; evasiveness about first- and second-year attrition.

Go deeper — further reading

- **Engineering hiring trends and workforce planning, 2026** — the source of the three-jobs-per-candidate figure, and the clearest account of where demand is concentrated by discipline and geography.

- **Engineering Jobs, Class of 2026 and Beyond** — starting salaries, BLS medians and growth projections by discipline in one place, with the demographic picture: nearly half of working US engineers are 50 or over.

- **Is Electrical Engineering a Good Career in 2026** — on the data-centre demand story, including the Lawrence Berkeley National Laboratory projection that AI data centres could consume 6–12% of US electricity by 2028.

- **PE License Salary: How Much More Do PEs Make?** — the licensure premium in numbers, drawing on BLS wage data and NSPE salary surveys.

The counterargument — read this one:

> The strongest case against our score isn't that engineering is more exposed than we say. It's that we may be crediting the licence with protection that is really being provided by demographics.
>
> Three open positions per candidate and an ageing workforce would produce a strong graduate market with or without the PE stamp. If the retirement wave passes and the pipeline refills, a reader might reasonably ask whether the structural protection was ever doing the work we attribute to it.
>
> **Where we land:** we think both are real and separable. The licence protects the *act* — no amount of workforce supply changes who may legally stamp a drawing. The demographics protect the *market* and will fade. So we'd expect engineering's score to hold even as hiring conditions normalise, and we'd treat a rising score alongside an easing labour market as evidence we were wrong about which factor mattered.

Bottom line

Engineering is the strongest technical bet in this index, and it is chronically under-chosen by exactly the students who would suit it.

For twenty years the default advice to a mathematically-able teenager was software. On the evidence, the same aptitude pointed at the physical world produces materially better protection and a materially better graduate market — 4.0 against 8.1 at entry, roughly three open positions per candidate against 6.1% unemployment.

The protection is real and structural. A legal monopoly on the stamp, personal liability, site work that cannot be done remotely, and a licence that mandates the supervised years that protect the on-ramp.

Two honest cautions. The desk-based end of the profession is drifting at nearly double the rate of the site-based end, so the discipline and the setting matter. And the logged-hours question in section 5 is a genuine open problem the profession has not yet confronted.

So the useful advice is: choose the discipline deliberately rather than defaulting to mechanical, aim at the site rather than the desk, take the placement, and start the PE track the day you graduate.

And for a student currently choosing computer science because they're good at maths and like building things — this is the comparison worth an hour of real conversation, not a passing mention.

Discussion questions

Part 1 — About the work itself

1. What made engineering appealing? Push past "I'm good at maths" to something specific. 2. Do they want to *design* things or *build* things? Those point to different disciplines and very different days. 3. Are they comfortable being responsible when something physical fails? That's the defining trait. 4. Do they like working inside constraints, or does that feel limiting? Engineering is problem-solving inside hard limits.

Part 2 — Reacting to the findings

5. Licensed engineering scores 4.0 and entry-level software 8.1. Does that change anything? 6. Roughly three open engineering positions per qualified candidate, against 6.1% CS graduate unemployment. Was that surprising? 7. Desk-based analysis scores 6.3 and site-based work 4.0. Does that affect which roles they'd aim for?

Part 2b — The harder conversation

8. Engineering has high attrition in the first two years, concentrated in maths-heavy foundations. How do they feel about that honestly? 9. Licensure takes about four supervised years after graduating. Have they pictured that timeline? 10. Site-based roles often mean travel or relocation early on. Is that appealing or a problem?

Part 2c — The other side

11. Of what engineers say is rewarding — building things that persist, working at scale, hard constraint problems, being trusted with public safety — which two matter most? 12. Looking at "who this work suits": which items sound like them? Answer separately, then compare.

Part 2d — The AI question, practically

13. The profile raises a question about logged licensure hours — whether supervising generated work builds the same judgment as producing it. What do they make of that? 14. Of the five capabilities — verify, specify, judge applicability, own the stamp, bridge to site — which appeals most?

Part 3 — Testing it against reality

15. **The most valuable single action:** find a working engineer and ask what the graduates on their team actually do, and how that differs from when they started. 16. Can they visit a site? A construction site, plant or substation tells you more in an afternoon than any brochure.

Part 4 — About the program

17. Is the programme accredited for professional licensure where they intend to practise? Check directly. 18. What proportion complete a placement, and does it include site work? 19. What's the first- and second-year attrition rate?

Part 5 — Widening the frame

20. **Have they compared this properly with computer science?** Same aptitude, very different position. This is the single most useful comparison in the index for a maths-able student. 21. Have they looked at construction management or the skilled trades? Both score better still, and both are chronically under-chosen. 22. Which discipline, and why? Electrical and civil are in the strongest demand positions right now.

Part 6 — For the parent, alone

23. Did I steer them toward computing? On what evidence, and is it still current? 24. Am I comfortable with a demanding degree that has real attrition? 25. What's the one thing I most want them to understand — in a sentence, without a number in it?

This is analysis and informed judgment about a fast-moving situation, not a guarantee or a prediction. For engineering specifically, we've flagged that some of the current strength in the graduate market is demographic and cyclical rather than structural.

Scores for 2023 and 2025 are retrospectively calculated using the current methodology. Scores measure exposure to what AI can already do — not how much any particular employer has deployed.

Appendix — technical scoring

The factor scores behind the headline number, shown for the **Civil / structural (PE, site-based)** track. Each factor is rated 0–10 and carries a fixed weight; *exposure* factors push the score up, *protection* factors pull it down. Full method at pivotum.ai/methodology.

FACTOR	WEIGHT	RATING	EFFECT
How much of this job can AI already do?	35%	6.5	**Exposure ↑**
How hard will it be to land that first job?	15%	3.0	**Protection ↓**
Does it have to be done in person, with your hands?	15%	6.0	**Protection ↓**
Does someone need a human they can trust and hold responsible?	15%	7.5	**Protection ↓**
Does the law require a licensed human?	10%	9.0	**Protection ↓**
How often does the job hit genuinely new, high-stakes situations?	10%	8.0	**Protection ↓**

Scores run 0–10, where 10 is most exposed to what AI can already do. Re-scored every six months.